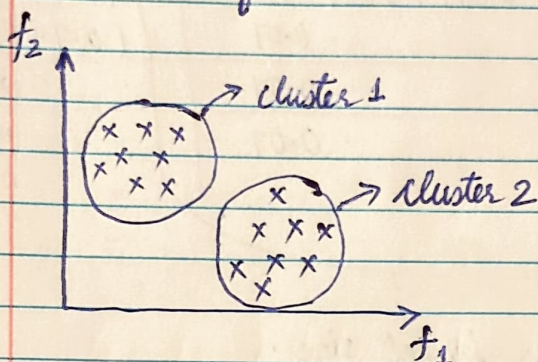


K means Clustering { unsupervised ML }
no specific o/p

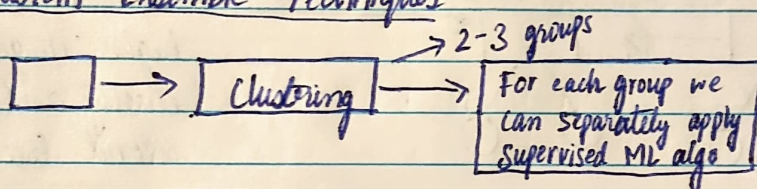
clusters
↓
Similar kind of data

o/p f_1 f_2
- -
- -
- -



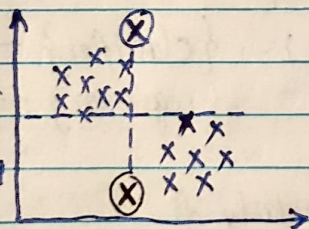
K means $\rightarrow$ K = centroids

Custom Ensemble Techniques

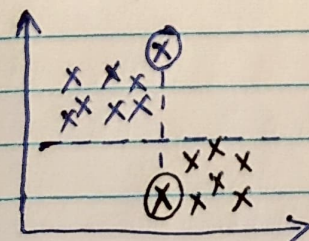


Steps to find in no of groups or K means

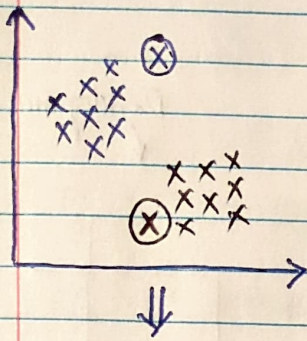
- 1) We try with different K values $\rightarrow$ centroids
Let say $K=2$, then



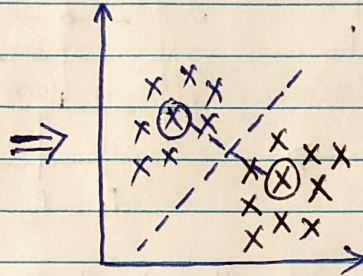
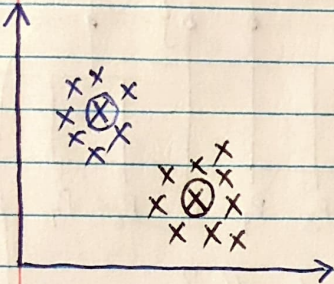
- 2) Initialize K number of centroids, I had initialized 2 centroids randomly in space
3) We use Euclidean distance b/w the centroids to find which points are near to the centroids



now the colour changes



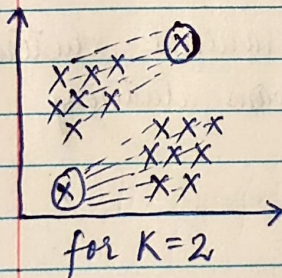
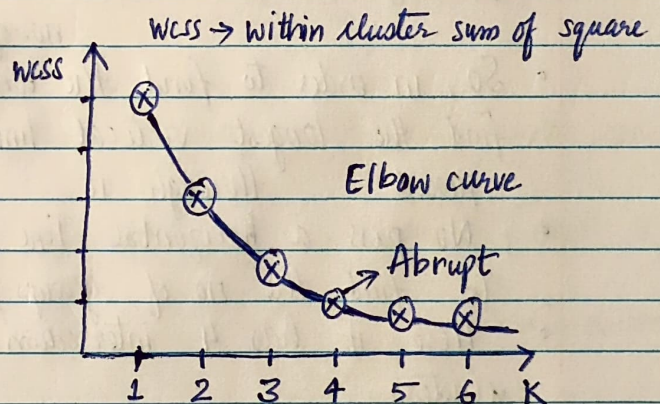
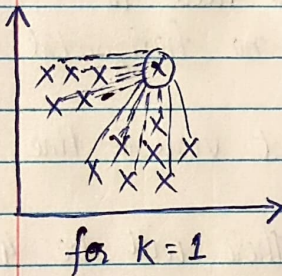
3) Compute the average of points of each group to update the centroid



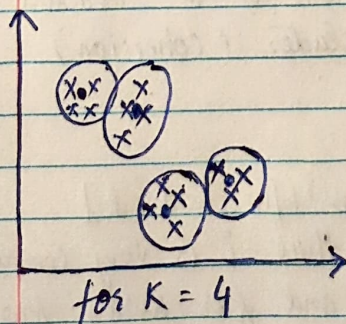
Now again the distance is calculated, now we get the perfect centroid and respective group

Elbow method (helps in finding optimized K value)

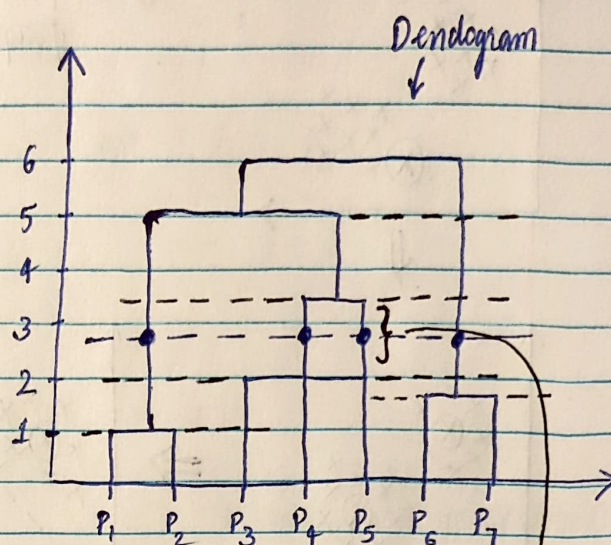
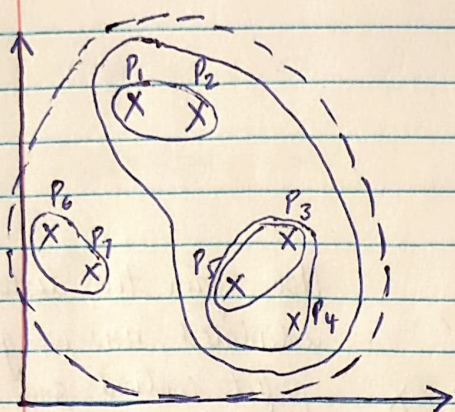
Let's say for $K=1, 10$



- Elbow method is only used for finding the K value
- For validating the model is performing well or not, we use Silhouette Score
- When there is an abrupt change in the Elbow curve we get the appropriate K value



Hierarchical Clustering



$P_1 - P_2$ { distance b/w them is shortest of all }
 $P_6 - P_7$ { 2nd lowest distance pair }
 $P_5 - P_3$

longest
vertical line

- So, in order to find the longest line you need to find the longest vertical line that has no horizontal line passed through it.
- No pass a horizontal line through longest vertical line to find the no of groups/clusters.
- Here it has 4 intersection points so there will be 4 clusters
- If your dataset is small go with Hierarchical clustering as it takes more time than K means clustering and vice-versa.

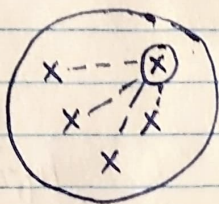
Silhouette Method

- The silhouette coefficient or silhouette score is a measure of how similar a data-point within a cluster (cohesion) compared to other clusters (separation)
- The value of the silhouette coefficient is b/w $[-1, 1]$
- The score of 1 denotes the best, meaning that it is very compact within the cluster to which it belongs and far away from the other clusters, the worst value is -1.
- Values near 0 denotes overlapping clusters.

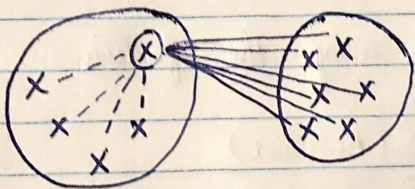
$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$b(i) \rightarrow$ average distance from i to all clusters to which i does not belong

$a(i) \rightarrow$ average distance b/w i and all other data points in the cluster to which i belongs.

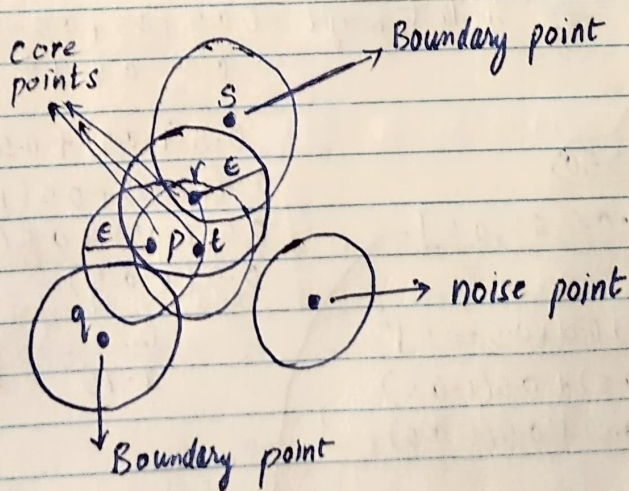


average of all the distance within cluster
↓
 $a(i)$

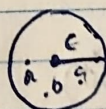


average of all the distance b/w i and all other datapoints outside of cluster.
↓
 $b(i)$

DBSCAN - Density based Spatial Clustering of Applications with noise



core point $\{ \min = 3 \}$



ϵ - radius distance